

Swarnim Varshneya

SOFTWARE ENGINEER | BACKEND SYSTEMS | PLATFORM ENGINEERING

New Delhi | svarshneyas@gmail.com | +91-9891344382 | linkedin.com/codeswav | github.com/swarmeme

Skills

Languages : Python, Go (Golang), SQL, Bash, React.js, Javascript (basic)

Tools and Platform : Git, Github, JIRA, Confluence, Linux

Cloud and Infrastructure : Kubernetes, CI/CD Pipelines, Docker, Google Cloud Platform (GCP), Prometheus, Grafana

Databases : MySQL, MongoDB

Backend : REST APIs, gRPC, Protobufs, Distributed Systems, Backend Services, Microservices

Core Concepts : Data Structures and Algorithms (DSA), Object Oriented Programming (OOP), pandas, scikit-learn, TensorFlow, Data Analysis, SwiftUI

Experience

Software Engineer - Backend (Application Engineer) • Flipkart – Bengaluru June 2024 – April 2026

- **Backend Development and Automation:** Built scalable backend services in Python and Go to automate DBaaS orchestration workflows, integrating REST APIs for cluster provisioning, client allocation, and lifecycle management, reducing manual provisioning effort by 98%.
- **Capacity Analytics System:** Developed a backend analytics module using Python, MySQL, and MongoDB to evaluate capacity utilization across 900+ production clusters, enabling migration readiness forecasting and detection of idle and over-provisioned resources. Reduced idle capacity by 60% and on call load by 20% to 30%.
- **Distributed Version Tracking and State Synchronization:** Modified VM level agent boot configurations to capture MySQL node version metadata and propagated updates via gRPC to Orchestrator and REST services to update in MongoDB documents. Enhanced internal React.js UI with a dedicated MySQL version visibility section to surface real time node state improving migration observability and failure diagnosis.
- **API Validation and Error Propagation:** Traced silent configuration failures across React UI and Golang REST services, refactored backend validation logic to ensure correct value propagation and structured error handling, and updated UI rendering logic to strictly reflect backend validated responses, eliminating silent configuration failures.
- **Distributed Systems Migration Engineering:** Built automated validation frameworks for MySQL 5.7 to 8.0 upgrades across 900+ clusters, ensuring zero downtime migrations and high system availability.
- **Performance Debugging and Data Consistency:** Resolved complex MySQL replication failures through binlog analysis and system level tracing, restoring cluster health and ensuring data consistency.

Data Science Intern • Punjab National Bank – New Delhi Feb 2024 – May 2024

- **Development of Machine Learning Models:** Led the development of a loan scheme classification model, reducing missing values to 10% and improving accuracy over the previous models using regression models. Implemented four regression models with scikit-learn and pandas on Jupyter Notebook, achieving a 10-20% reduction in null values via IBM Cloud Pak CPD pipelines.

Education

SRM Institute of Science and Technology | Bachelors of Technology in Computer Science and Engineering Sept 2020 – May 2024
CGPA: 9.2/10

Certifications

Google Cloud Fundamentals: Core Infrastructure Google
Introduction to iOS Mobile Application Development Meta
Data Mining NPTEL

Projects

Club Membership Churn Prediction | Python, pandas, scikit-learn, seaborn Nov 2023

- Engineered data analysis and preprocessing using pandas, scikit-learn, and seaborn to train a Random Forest Classifier for predicting membership cancellations. Conducted Exploratory Data Analysis (EDA) to identify cancellation patterns, achieving 66.12% accuracy and 82.20% precision.

Restaurant Reservation System | Swift and SwiftUI. Sept 2023

- Designed a Restaurant Reservation User Interface in SwiftUI, focusing on layout design and SwiftUI concepts.